



Subject	Computer	worksheet	
Student's Name		Grade	8

Q1: choose the correct answer:

What is the correct way to declare a variable in JavaScript?			
var x;	int x;	double x;	char x;
Which data type is used to represent text in JavaScript?			
Number	String	Boolean	Array
Which method is used to create a new Promise object?			
new Promise()	Promise.create()	Promise.new()	Promise.make()
What is the correct way to handle errors in a Promise?			
catch()	error()	fail()	reject()
Which sorting algorithm has the best average-case time complexity?			
Bubble Sort	Insertion Sort	Quick Sort	Selection Sort

Q2 : Read and put True(✓) or false (✗) :

- 1) JavaScript is a case-sensitive language. (✓)
- 2) JavaScript is a synchronous language by default. (✓)
- 3) JavaScript is a compiled language. (✗)
- 4) The typeof operator returns the data type of a variable. (✓)



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Q1: What are Data Types in JavaScript?

- Numbers
- Strings
- Boolean
- Symbol
- Objects
- Functions
- Arrays

Q2: Which symbol is used for comments in JavaScript?

- single-line comment.
`// Single line comment`
- multi-line comment
`/*
Multi-line comments
...
*/`



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Q1 : Read and put True(✓) or false (✗) :

1. An algorithm must always terminate after a finite number of steps. (✓)
2. Merge Sort is an example of a divide-and-conquer algorithm. (✓)
3. Dynamic programming is used to solve problems with overlapping subproblems. (✓)
4. Dijkstra's algorithm can be used to find the shortest path in a graph with negative weights. (✗)
5. The space complexity of an algorithm measures the amount of memory it uses. (✓)

Q2 : Arrange the following steps of the Merge Sort algorithm in the correct order:

- (3) Merge the two sorted halves.
- (2) Recursively sort the left half.
- (4) Divide the array into two halves.
- (1) Recursively sort the right half.



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Q1: choose the correct answer:

1. Which of the following is NOT a characteristic of an algorithm?

- a) Definiteness
- b) Finiteness
- c) Efficiency
- d) Ambiguity

2. What is the time complexity of binary search in the worst case?

- a) $O(1)$
- b) $O(\log n)$
- c) $O(n)$
- d) $O(n^2)$

3. Which sorting algorithm has the best average-case time complexity?

- a) Bubble Sort
- b) Quick Sort
- c) Insertion Sort
- d) Selection Sort



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4. In the context of algorithms, what does "greedy" mean?
- a) Making the best local choice at each step
 - b) Exploring all possible solutions
 - c) Dividing the problem into smaller subproblems
 - d) Using recursion to solve problems
5. Which data structure is typically used for implementing a breadth-first search (BFS)?
- a) Stack
 - b) Queue
 - c) Priority Queue
 - d) Linked List



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Match the word in column (A) with the correct option
in column (B):

Attribute	#	value
1- Color	3	<table>
2- create a hyperlink	1	#123456
3- create a table	4	
4- insert an image	2	<a>

Arrange the following steps in the correct order to create and call a JavaScript function.

- (4) Call the function.
- (1) Define the function using the function keyword.
- (2) Add parameters to the function (if needed).
- (3) Write the function body.



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Find the output of the following code:

```
let x = 5;  
let y = 2;  
let z = x + y;
```

output7

```
let x = 5;  
let y = 2;  
let z = x - y;
```

output3